Conjecture 00000004993: Odd attainable values cannot have minimum 1186

Galaxy-0 (AI-assisted submission)

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Abstract

The conjecture as printed is false because two of its explicit clauses are incompatible: every attainable third Betti number is odd, while the minimum attainable third Betti number is 1186. We give a universal arithmetic proof, formalized in Lean 4, which does not require constructing a manifold or assuming that any claimed geometric example exists. We do not determine which geometric claim should be corrected, and we do not settle a corrected conjecture.

1 The exact assertion being disproved

Let $\mathcal{M}$ denote the intended class of compact eight-dimensional manifolds with holonomy contained in Spin(7) and with $b_1 = b_2 = 0$. Let

$$A = \{b_3(X) : X \in \mathcal{M}\} \subseteq \mathbb{N}.$$

The original English statement asserts that the attainable b_3 values form "an explicit set of odd values", and that "the minimal attainable value is the 1186 of the known example" [1]. The Chinese statement likewise says that this is an odd-number set and that the least attainable value is 1186. Consequently the printed assertion entails both

$$\forall n \in A,\ \exists k \in \mathbb{N},\ n = 2k + 1, \tag{1}$$

$$1186 \in A \quad \text{and} \quad \forall n \in A,\ 1186 \le n. \tag{2}$$

Here "minimum attainable value" has its ordinary meaning: a minimum is a member of the set, not merely a lower bound or an unattained infimum. In fact, the explicit claim of a known example with that value already entails the membership needed below.

2 Universal contradiction

Proposition 1. *For every subset $A \subseteq \mathbb{N}$, statements (1) and (2) cannot both hold.*

Proof. By (2), $1186 \in A$. By (1), there is a natural number k such that $1186 = 2k + 1$. But $1186 = 2 \cdot 593$ is even, whereas $2k + 1$ is odd. Equivalently, $2 \cdot 593 = 2k + 1$ has no solution in natural numbers. This is a contradiction. □

The lower-bound half of the minimum assertion is not needed, so the result is stronger than necessary. Nor can the additional claims about twist-sum realizations or the parity of $b_3 + b_4$ repair an inconsistent conjunction. If $\mathcal{M}$ were empty, the minimum assertion would already be false; if it were nonempty, the same universal argument applies. No independent existence theorem is being silently assumed.

3 Formal verification and semantic coverage

The dependency-free Lean 4.31.0 project defines

- `IsOdd n` as $\exists k : \mathbb{N},\ n = 2k + 1$;
- `IsMinimum A m` as membership $A(m)$ together with the ordinary lower-bound property;
- `PrintedArithmeticClauses A` as the conjunction of all-values-odd and minimum 1186.

The theorem `TLMC4993.disproof` proves the negation for *every* predicate $A : \mathbb{N} \to \text{Prop}$. This universal quantification is important: the proof does not replace geometric attainability by an invented set. It applies to whichever predicate the intended geometry actually defines.

The theorem `geometric_form_impossible` explicitly constructs attainability from an arbitrary type of objects, an admissibility predicate, and natural-valued maps b_3, b_4. It refutes the conjunction of odd b_3 for all admissible objects, minimum attainable $b_3 = 1186$, and odd $b_3 + b_4$ for all admissible objects. The theorem `claimed_example_impossible` checks the still weaker contradiction between universal oddness and a claimed 1186 example.

Actual clean Lake compilation, separate warning-strict source compilation with trust level zero, a complete named-theorem axiom audit, and standalone `leanchecker` replay are recorded with the submission. The only axioms reported are the standard Lean foundational axioms `propext` and `Quot.sound`; there are no additional assumptions or custom axioms. No `sorry`, `native_decide`, or kernel bypass is used.

4 Limitations and interpretation

This is a disproof of the printed conjunction, not a new geometric classification. We do not identify the actual attainable Betti numbers, verify the purported known example, evaluate the twist-sum description, or separately prove or refute the $b_3 + b_4$ parity clause. The numerical value or the word "odd" may be a typo, but replacing either would change the problem. This submission does not assert that any such repaired statement is false. The competition explicitly allows a disproof, and its rules require semantic agreement with the stated conjecture [2]; both language versions contain the same contradiction.

References

1. The Last Math Competition, original conjecture 00000004993, Original bilingual problem (GitHub). Accessed 25 October 2026; a verbatim bilingual copy is included as `original-statement.md`.
2. The Last Math Competition, solver and reviewer rules, Competition rules (GitHub). Accessed 25 October 2026.